

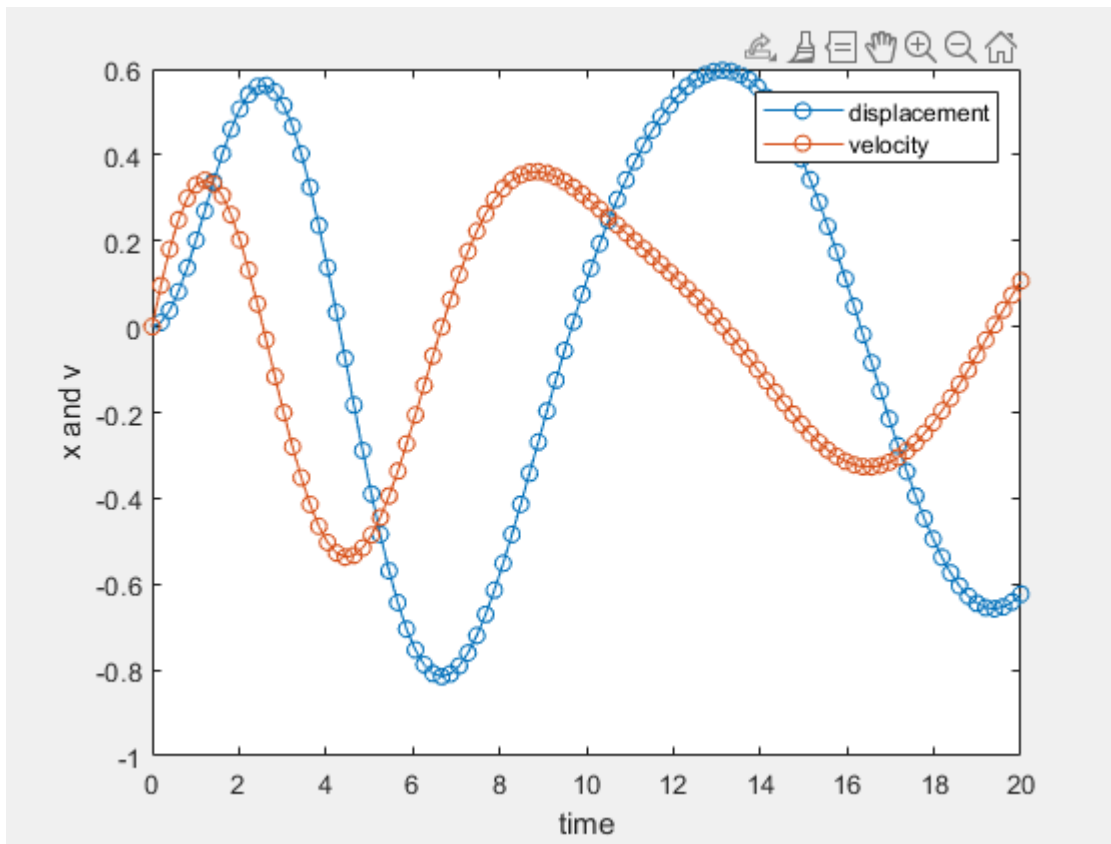
Express the given second-order ODE as a pair of first-order ODEs:

Let $y_1 = x$, and $y_2 = \frac{dy_1}{dt} \rightarrow \frac{dy_2}{dt} = \frac{d^2y_1}{dt^2} = \frac{d^2x}{dt^2}$, then we have the following ODE system:

$$\frac{dy_1}{dt} = y_2$$

$$\frac{dy_2}{dt} = \frac{1}{m} * (P \cos(wt) - c y_2 - k_1 y_1 - k_3 y_1^3)$$

```
init=[0 0]; % initial condition
t=linspace(0,20);
ode45(@myfunc,t,init);
legend('displacement','velocity')
xlabel('time')
ylabel('x and v')
```



```
function [dydt] = myfunc(t,y)
m=1;c=0.4;P=0.5;w=0.5;k1=1;k3=0;
dydt = zeros(2,1);
dydt(1) = y(2);
dydt(2) = (1/m)*(P*cos(w*t)-c*y(2)-k1*y(1)-k3*y(1)^3);
end
```